

PINNs for optimal control

Ngày 15 tháng 7 năm 2026

1 Problem formulation

Let $\Omega \subset \mathbb{R}^d$ ($d = 1, 2$) be a bounded domain with sufficiently smooth boundary $\partial\Omega$, let $T > 0$, and set $Q := \Omega \times (0, T)$ and $\Sigma := \partial\Omega \times (0, T)$. We seek a state $y : Q \rightarrow \mathbb{R}$ and a Dirichlet boundary control $u : \Sigma \rightarrow \mathbb{R}$ minimizing

$$J(y, u) = \frac{1}{2} \int_Q |y - y_d|^2 dx dt + \frac{\alpha}{2} \int_\Sigma |u|^2 ds dt, \quad (1)$$

subject to

$$\begin{cases} \partial_t y - \nu \Delta y + f(y) = g, & \text{in } Q, \\ y = u, & \text{on } \Sigma, \\ y(x, 0) = y_0(x), & \text{in } \Omega. \end{cases} \quad (2)$$

Here $\nu > 0$ is the diffusion coefficient, g is a prescribed source term, $y_d \in L^2(Q)$ is the desired trajectory, and $\alpha > 0$ is the Tikhonov regularization parameter. The source g is fixed problem data and is not an optimization variable. The case $g = 0$ is included. The boundary condition is of Dirichlet type, so the control acts directly on the boundary trace of the state.

2 Methodology

We compare a direct PINN, which minimizes a penalized form of the constrained objective, with an indirect PINN, which solves the first-order optimality system. Both methods use a single control network $U_\psi(x, t)$.

2.1 Direct PINN Formulation

The state and control are approximated by $Y_\theta(x, t)$ and $U_\psi(x, t)$. Define

$$r_y(x, t; \theta) := \partial_t Y_\theta - \nu \Delta Y_\theta + f(Y_\theta) - g. \quad (3)$$

The direct loss is

$$\mathcal{L}_{\text{dir}} = w_{\text{res}} \mathcal{L}_{\text{res}}^y + w_{\text{bc}} \mathcal{L}_{\text{bc}}^y + w_{\text{ic}} \mathcal{L}_{\text{ic}}^y + \frac{1}{2} \mathcal{L}_{\text{track}} + \frac{\alpha}{2} \mathcal{L}_{\text{reg}}, \quad (4)$$

where

$$\mathcal{L}_{\text{res}}^y := \frac{1}{N_{\text{int}}} \sum_{j=1}^{N_{\text{int}}} |r_y(x_j, t_j; \theta)|^2, \quad (5)$$

$$\mathcal{L}_{\text{bc}}^y := \frac{1}{N_{\text{bc}}} \sum_{j=1}^{N_{\text{bc}}} |Y_\theta(x_j^b, t_j^b) - U_\psi(x_j^b, t_j^b)|^2, \quad (6)$$

$$\mathcal{L}_{\text{ic}}^y := \frac{1}{N_{\text{ic}}} \sum_{i=1}^{N_{\text{ic}}} |Y_\theta(x_i, 0) - y_0(x_i)|^2, \quad (7)$$

$$\mathcal{L}_{\text{track}} := \frac{1}{N_{\text{int}}} \sum_{j=1}^{N_{\text{int}}} |Y_\theta(x_j, t_j) - y_d(x_j, t_j)|^2. \quad (8)$$

In one space dimension the boundary consists of two endpoints, and the control regularization uses counting measure on $\partial\Omega$:

$$\mathcal{L}_{\text{reg}} := \frac{1}{N_{\text{bc}}} \sum_{j=1}^{N_{\text{bc}}} (|U_\psi(0, t_j^b)|^2 + |U_\psi(1, t_j^b)|^2). \quad (9)$$

This formulation is simple and avoids an adjoint network, but boundary satisfaction depends on the penalty weight and can be delicate [1].

2.2 Indirect PINN Formulation with Hard Constraints

2.2.1 First-order optimality system

With

$$\mathcal{L}(y, u, \lambda) = J(y, u) + \int_Q \lambda (\partial_t y - \nu \Delta y + f(y) - g) \, dx \, dt, \quad (10)$$

integration by parts gives

$$\partial_t y - \nu \Delta y + f(y) - g = 0 \quad \text{in } Q, \quad (11)$$

$$-\partial_t \lambda - \nu \Delta \lambda + f'(y) \lambda + (y - y_d) = 0 \quad \text{in } Q, \quad (12)$$

$$\alpha u + \nu \partial_n \lambda = 0 \quad \text{on } \Sigma, \quad (13)$$

$$y = u, \quad \lambda = 0 \quad \text{on } \Sigma, \quad (14)$$

$$y(x, 0) = y_0(x), \quad \lambda(x, T) = 0 \quad \text{in } \Omega. \quad (15)$$

For $\Omega = (0, 1)$, $\partial_n \lambda(0, t) = -\lambda_x(0, t)$ and $\partial_n \lambda(1, t) = \lambda_x(1, t)$.

2.2.2 One control network and hard constraints

We use one network $U_\psi(x, t)$ and define its two boundary traces by

$$u_0(t) := U_\psi(0, t), \quad u_1(t) := U_\psi(1, t). \quad (16)$$

For $\Omega = (0, 1)$, the boundary lifting and state trial function are

$$B_\psi(x, t) := (1 - x)U_\psi(0, t) + xU_\psi(1, t), \quad (17)$$

$$Y_\theta(x, t) := B_\psi(x, t) + x(1 - x)N_\theta(x, t). \quad (18)$$

Thus $Y_\theta(0, t) = U_\psi(0, t)$ and $Y_\theta(1, t) = U_\psi(1, t)$ for every parameter value. The adjoint is parameterized as

$$\Lambda_\phi(x, t) := x(1 - x)(T - t)M_\phi(x, t), \quad (19)$$

which enforces both $\Lambda_\phi = 0$ on the spatial boundary and $\Lambda_\phi(x, T) = 0$. The state initial condition remains a soft constraint.

2.2.3 Indirect PINN loss

Define

$$R_y := \partial_t Y_\theta - \nu \Delta Y_\theta + f(Y_\theta) - g, \quad (20)$$

$$R_\lambda := -\partial_t \Lambda_\phi - \nu \Delta \Lambda_\phi + f'(Y_\theta) \Lambda_\phi + Y_\theta - y_d, \quad (21)$$

$$R_{u,0} := \alpha U_\psi(0, t) - \nu \partial_x \Lambda_\phi(0, t), \quad (22)$$

$$R_{u,1} := \alpha U_\psi(1, t) + \nu \partial_x \Lambda_\phi(1, t). \quad (23)$$

The indirect loss is

$$\mathcal{L}_{\text{ind}} = w_y \mathcal{L}_y + w_\lambda \mathcal{L}_\lambda + w_{\text{st}} \mathcal{L}_{\text{st}} + w_{\text{ic}} \mathcal{L}_{\text{ic}}, \quad (24)$$

with

$$\mathcal{L}_y := \frac{1}{N_{\text{int}}} \sum_j |R_y(x_j, t_j)|^2, \quad (25)$$

$$\mathcal{L}_\lambda := \frac{1}{N_{\text{int}}} \sum_j |R_\lambda(x_j, t_j)|^2, \quad (26)$$

$$\mathcal{L}_{\text{st}} := \frac{1}{2} \left(\frac{1}{N_{\text{bc}}} \sum_j |R_{u,0}(t_j^b)|^2 + \frac{1}{N_{\text{bc}}} \sum_j |R_{u,1}(t_j^b)|^2 \right), \quad (27)$$

$$\mathcal{L}_{\text{ic}} := \frac{1}{N_{\text{ic}}} \sum_i |Y_\theta(x_i, 0) - y_0(x_i)|^2. \quad (28)$$

There are no state-boundary, adjoint-boundary, or adjoint-terminal penalties because these constraints are satisfied by construction.

3 Error Analysis

In this section, we establish an error estimation framework for the proposed Indirect PINN method with hard constraint. We decompose the total error into several components, derive a conditional a priori error bound, and construct a computable a posteriori error indicator that is consistent with the meshfree, collocation-based nature of PINNs. Unlike finite-element residual estimators, our indicator does not rely on a computational mesh and therefore contains no inter-element jump terms.

3.1 The Optimality System and Its Residuals

Recall the first-order optimality system for our problem: find (y, u, λ) such that

$$\partial_t y - \nu \Delta y + f(y) - g = 0, \quad \text{in } Q, \quad (29)$$

$$-\partial_t \lambda - \nu \Delta \lambda + f'(y)\lambda + (y - y_d) = 0, \quad \text{in } Q, \quad (30)$$

$$\alpha u + \nu \partial_n \lambda = 0, \quad \text{on } \Sigma, \quad (31)$$

$$y = u, \quad \text{on } \Sigma, \quad (32)$$

$$\lambda = 0, \quad \text{on } \Sigma, \quad (33)$$

$$y(x, 0) = y_0(x), \quad \text{in } \Omega, \quad (34)$$

$$\lambda(x, T) = 0, \quad \text{in } \Omega. \quad (35)$$

This is precisely the system (11)–(33) of Section 2, and we use the same constant α as in the cost functional (1) (we no longer introduce the separate symbols β_Q, β_R used in an earlier draft, to keep notation consistent across sections; the tracking term always carries weight 1 and the control term always carries weight α).

Let $Y_\theta, U_\psi, \Lambda_\phi$ be the neural network approximations built with the hard constraints of Section 2.2.2. The state and adjoint boundary conditions and the adjoint terminal condition hold exactly, whereas the state initial condition is penalized. We define

$$\mathcal{R}_y(x, t; \theta, \psi) := \partial_t Y_\theta - \nu \Delta Y_\theta + f(Y_\theta) - g, \quad (36)$$

$$\mathcal{R}_\lambda(x, t; \theta, \phi) := -\partial_t \Lambda_\phi - \nu \Delta \Lambda_\phi + f'(Y_\theta)\Lambda_\phi + (Y_\theta - y_d), \quad (37)$$

$$\mathcal{R}_u(x, t; \psi, \phi) := \alpha U_\psi + \nu \partial_n \Lambda_\phi, \quad (x, t) \in \Sigma. \quad (38)$$

Because the boundary and adjoint terminal conditions are enforced by construction, only the three optimality residuals and the state initial mismatch enter the loss:

$$\mathcal{L}_{\text{ind}}(\theta, \psi, \phi) = w_y \mathcal{L}_y + w_\lambda \mathcal{L}_\lambda + w_u \mathcal{L}_u + w_{\text{ic}} \mathcal{L}_{\text{ic}}, \quad (39)$$

where

$$\mathcal{L}_y := \frac{1}{N_{\text{int}}} \sum_{j=1}^{N_{\text{int}}} |\mathcal{R}_y(x_j, t_j; \theta, \psi)|^2, \quad (40)$$

$$\mathcal{L}_\lambda := \frac{1}{N_{\text{int}}} \sum_{j=1}^{N_{\text{int}}} |\mathcal{R}_\lambda(x_j, t_j; \theta, \phi)|^2, \quad (41)$$

$$\mathcal{L}_u := \frac{1}{N_{\text{bc}}} \sum_{j=1}^{N_{\text{bc}}} |\mathcal{R}_u(x_j^b, t_j^b; \psi, \phi)|^2, \quad (42)$$

$$\mathcal{L}_{\text{ic}} := \frac{1}{N_{\text{ic}}} \sum_{i=1}^{N_{\text{ic}}} |Y_\theta(x_i, 0) - y_0(x_i)|^2. \quad (43)$$

Each $\mathcal{L}_\bullet$ is a Monte-Carlo (or quasi-Monte-Carlo) quadrature approximation of a continuous L^2 -norm of the corresponding residual, e.g. $\mathcal{L}_y \approx \|\mathcal{R}_y\|_{L^2(Q)}^2$; this correspondence is what links the *discrete, trainable* loss to the *continuous* PDE residuals used in the estimates below.

3.2 Standing Assumptions

The estimates in this section are conditional on the following standing assumptions, which we state explicitly rather than leaving implicit.

- (A1) **Well-posedness.** The optimality system (29)–(33) admits a unique solution (y^*, u^*, λ^*) in an appropriate energy class, and the linearization of this system about (y^*, λ^*) satisfies an inf-sup (stability) condition with constant C_{stab} .
- (A2) **Regularity of the Dirichlet control problem.** Dirichlet boundary control of a parabolic equation is, in general, a low-regularity problem: for boundary data $u \in L^2(\Sigma)$ the state y is typically only a *very weak solution* (transposition solution), and the trace $\partial_n \lambda|_\Sigma$ appearing in (31) need not exist in the classical sense [2, 3]. We assume throughout that the data (y_0, y_d, u^*) are smooth enough, and Ω is regular enough, that (y^*, λ^*) lie in $H^{2,1}(Q)$ -type spaces in which all traces below are classically well defined. This is a genuine restriction, not a technicality; results that require only $u \in L^2(\Sigma)$ are outside the scope of the estimates below.
- (A3) **Nonlinearity.** $f \in C^1(\mathbb{R})$ and f' is bounded below, $f'(s) \geq -c_f$ for some $c_f \geq 0$ (satisfied by the Allen–Cahn nonlinearity $f(y) = y^3 - y$, for which $f'(s) = 3s^2 - 1 \geq -1$).
- (A4) **Approximation capacity.** The network architectures for $Y_\theta, U_\psi, \Lambda_\phi$ are rich enough that the best-approximation error ϵ_{net} defined in (46) can be made arbitrarily small by increasing width/depth. No explicit rate in network size is claimed here; this is treated as an assumption, not a proven property of the architecture.

3.3 Decomposition of the Total Error

Let $e_y := y^* - Y_\theta$, $e_\lambda := \lambda^* - \Lambda_\phi$, $e_u := u^* - U_\psi$ denote the errors in the state, adjoint, and control, relative to the exact solution of the optimality system (not to a discretized/mesh solution). The total error is measured in the norm

$$\|(e_y, e_\lambda, e_u)\|_{\mathcal{E}}^2 := \|e_y\|_{L^2(0,T;H^1(\Omega))}^2 + \|e_\lambda\|_{L^2(0,T;H^1(\Omega))}^2 + \|e_u\|_{L^2(\Sigma)}^2. \quad (44)$$

Using the triangle inequality and the stability assumption (A1), the total error can be decomposed into three main contributions:

$$\|(e_y, e_\lambda, e_u)\|_{\mathcal{E}} \lesssim \mathcal{E}_{\text{approx}} + \mathcal{E}_{\text{opt}} + \mathcal{E}_{\text{quad}}, \quad (45)$$

where:

- $\mathcal{E}_{\text{approx}}$: best-approximation error of the neural network architecture (item (A4)),
- $\mathcal{E}_{\text{opt}}$: optimization error, i.e. how far the trained loss (39) is from its infimum over the chosen architecture,

- $\mathcal{E}_{\text{quad}}$: quadrature (sampling) error between the Monte-Carlo loss $\mathcal{L}_{\text{ind}}$ and the continuous residual norms it estimates.

We do not list a separate boundary or terminal contribution because these conditions are hard-constrained. The state initial mismatch is retained explicitly in the trainable loss and in the estimator below.

3.4 A Priori Error Estimate

Theorem 1 (Conditional a priori estimate). *Under Assumptions (A1)–(A3), let (y^*, u^*, λ^*) be the exact solution of (29)–(33). Suppose the networks Y_θ , U_ψ , Λ_ϕ can approximate (y^*, λ^*, u^*) to accuracy ϵ_{net} in the $\mathcal{E}$ -norm,*

$$\inf_{\theta, \psi, \phi} \|(y^* - Y_\theta, \lambda^* - \Lambda_\phi, u^* - U_\psi)\|_{\mathcal{E}} \leq \epsilon_{\text{net}}, \quad (46)$$

and suppose training reaches parameters $(\theta^*, \psi^*, \phi^*)$ with continuous residual norm

$$\|\mathcal{R}_y\|_{L^2(Q)}^2 + \|\mathcal{R}_\lambda\|_{L^2(Q)}^2 + \|\mathcal{R}_u\|_{L^2(\Sigma)}^2 \leq \epsilon_{\text{loss}}^2. \quad (47)$$

Then

$$\|(y^* - Y_{\theta^*}, \lambda^* - \Lambda_{\phi^*}, u^* - U_{\psi^*})\|_{\mathcal{E}} \lesssim \epsilon_{\text{net}} + C_{\text{stab}} \epsilon_{\text{loss}}, \quad (48)$$

with C_{stab} the stability constant of Assumption (A1).

Sketch of proof. Write $e = (e_y, e_\lambda, e_u)$ for the error at the trained parameters, and split $e = e^{\text{approx}} + e^{\text{res}}$, where e^{approx} is the error of the *best* network approximation (controlled by ϵ_{net} via (A4)) and $e^{\text{res}} := (y^* - Y_{\theta^*}, \dots) - e^{\text{approx}}$ is what remains. By linearity, e^{res} solves the linearization of (29)–(31) about (y^*, λ^*) with right-hand side equal to $(\mathcal{R}_y, \mathcal{R}_\lambda, \mathcal{R}_u)$ evaluated at the trained networks (this uses (A2) to make the linearization and its traces meaningful, and (A3) to control the sign of the reaction term $f'(y^*)$ so that the linearized adjoint operator is coercive up to a lower-order term, via a standard exponential-weight / Gårding-inequality argument). The stability assumption (A1) then gives $\|e^{\text{res}}\|_{\mathcal{E}} \leq C_{\text{stab}} (\|\mathcal{R}_y\|_{L^2(Q)} + \|\mathcal{R}_\lambda\|_{L^2(Q)} + \|\mathcal{R}_u\|_{L^2(\Sigma)}) \leq C_{\text{stab}} \epsilon_{\text{loss}}$. Combining with $\|e^{\text{approx}}\|_{\mathcal{E}} \leq \epsilon_{\text{net}}$ and the triangle inequality gives (48). We emphasize that this is a *sketch*: a full proof requires making (A1)–(A2) precise (function spaces, trace theorems) for the specific f and domain considered, which is not carried out here. $\square$

Remark 1. Bound (48) uses the continuous residual norm ϵ_{loss} , not the Monte-Carlo loss $\mathcal{L}_{\text{ind}}$ that is actually minimized during training. The gap between the two is exactly $\mathcal{E}_{\text{quad}}$ in (45); see Section 3.6 for how it is controlled in practice.

Remark 2. For the Allen–Cahn nonlinearity $f(y) = y^3 - y$, $f'(s) = 3s^2 - 1 \geq -1$ is bounded below, which is what makes the coercivity argument in the proof sketch available; a genuine coercivity estimate for the linearized adjoint operator additionally requires either a smallness condition on T or an exponential time-weight $e^{-c_f t}$ to absorb the -1 reaction term, which we do not carry out in detail here.

3.5 A Posteriori Error Estimation

Because $Y_\theta, U_\psi, \Lambda_\phi$ are smooth functions on all of $\bar{Q}$ (for smooth activation functions), there is no computational mesh and no notion of an inter-element jump in the normal derivative — such jumps are identically zero for a globally differentiable network. We therefore build the a posteriori indicator directly from pointwise residual evaluation on a *verification point set*, distinct from (and typically much larger/denser than) the training collocation set, rather than from a finite-element-style mesh.

3.5.1 Verification-Point Residual Indicator

Let $\hat{Q} = \{(x_k, t_k)\}_{k=1}^K \subset Q$ and $\hat{\Sigma} = \{(x_k^b, t_k^b)\}_{k=1}^{K_b} \subset \Sigma$ be dense verification point sets (e.g. a fixed low-discrepancy or grid sample, held out from training) with associated quadrature weights $\{\omega_k\}$, $\{\omega_k^b\}$ summing to $|Q|$ and $|\Sigma|$ respectively. Define the global a posteriori indicator

$$\eta^2 := \sum_{k=1}^K \omega_k |\mathcal{R}_y(x_k, t_k)|^2 + \sum_{k=1}^K \omega_k |\mathcal{R}_\lambda(x_k, t_k)|^2 + \sum_{k=1}^{K_b} \omega_k^b |\mathcal{R}_u(x_k^b, t_k^b)|^2 + \eta_{\text{ic}}^2, \quad (49)$$

where η_{ic}^2 approximates $\|Y_\theta(\cdot, 0) - y_0\|_{L^2(\Omega)}^2$. The adjoint terminal mismatch is identically zero by (19). Thus η is evaluated on points independent of training and approximates the continuous residual quantity entering Theorem 1. A post-hoc partition may be used only to visualize local contributions; it introduces no jump terms.

3.5.2 Reliability

Theorem 2 (Reliability). *Under Assumptions (A1)–(A3), and with η as in (49) computed on a sufficiently fine verification set,*

$$\|(e_y, e_\lambda, e_u)\|_{\mathcal{E}} \lesssim \epsilon_{\text{net}} + C_{\text{stab}}(\eta + \mathcal{E}_{\text{quad}}), \quad (50)$$

where $\mathcal{E}_{\text{quad}}$ is the (typically negligible, if K, K_b are large) quadrature error of (49) itself relative to the exact L^2 residual norms.

Sketch of proof. This follows conditionally from Theorem 1 by identifying η as a quadrature approximation of the continuous residual norm and retaining both the best-approximation and verification-quadrature contributions. $\square$

3.5.3 On "Efficiency" (Lower Bound)

A finite-element a posteriori estimator is typically also *efficient*: $\eta(E) \lesssim \|e\|_{1, D_E}$ up to data oscillation, proved with element-wise bubble functions supported on individual mesh cells. This construction relies essentially on having a mesh with compactly supported, element-local test functions. In the present meshfree PINN setting there is no such local test-function machinery available in general, and we are not aware of a proof of a matching lower bound for collocation-based PINN residual indicators of this type. We therefore do *not* claim an efficiency theorem here; (49) should be understood as a reliable (upper-bounding, up to constants) indicator only, useful for monitoring convergence and flagging poorly-resolved regions of Q , but not as a certified two-sided estimator. Establishing efficiency rigorously in the meshfree setting is left as an open direction.

3.6 Loss Function as an Error Indicator

Combining Theorem 1 with the identification of η as a quadrature estimate of ϵ_{loss} , we have informally

$$\|(e_y, e_\lambda, e_u)\|_{\mathcal{E}}^2 \lesssim \mathcal{L}_{\text{ind}} + \mathcal{E}_{\text{approx}} + \mathcal{E}_{\text{quad}}, \quad (51)$$

so a small training loss is *necessary but not sufficient* for a small true error: it must be accompanied by (i) a sufficiently expressive architecture ($\mathcal{E}_{\text{approx}}$ small) and (ii) a sufficiently dense/well-distributed collocation set ($\mathcal{E}_{\text{quad}}$ small). In particular, a low training loss evaluated *on the training points themselves* can be misleading if the network overfits those points; this is precisely why η in Section 3.5 is evaluated on an independent verification set.

3.7 Practical Implementation of the Error Estimator

The a posteriori indicator is computed after training as follows:

1. Sample a verification point set $\widehat{Q} \cup \widehat{\Sigma}$, independent of the training collocation points (e.g. a fresh random draw, or a fixed low-discrepancy grid), together with quadrature weights.
2. Evaluate $Y_\theta, U_\psi, \Lambda_\phi$ and their required derivatives at these points via automatic differentiation, and compute the residuals $\mathcal{R}_y, \mathcal{R}_\lambda, \mathcal{R}_u$.
3. Form the weighted sum (49).
4. (Optional, diagnostic only) Bin the pointwise squared residuals over a post-hoc space-time partition purely for visualization of where the error is concentrated; this partition does not enter the estimator's value.

The computational cost is comparable to one evaluation of the loss function on $K + K_b$ points, i.e. negligible relative to training.

3.8 Numerical Validation Strategy

To validate the error indicator, we propose the following procedure:

1. Construct a reference solution $(y_{\text{ref}}, u_{\text{ref}}, \lambda_{\text{ref}})$ with a fine-grid FEM/adjoint solver.
2. Train the Indirect PINN with an increasing number of training collocation points $N = N_{\text{int}} = N_{\text{bc}}$ (there is no mesh size h in the meshfree setting; N plays the analogous role).
3. Compute the true error $\|e\|_{\mathcal{E}}$ against the reference solution.
4. Compute the indicator η from (49) on a verification set independent of both the N training points and the reference-solution grid.
5. Compare η with the true error as N increases, and report the empirical ratio $\eta/\|e\|_{\mathcal{E}}$ (an *effectivity index*) rather than assuming a fixed algebraic rate; unlike FEM, PINN convergence in N is not generally guaranteed to follow a clean power law, and any observed rate should be reported empirically rather than asserted a priori.

This validates the reliability of the estimator in the sense of Theorem 2; it does not, and is not intended to, validate an efficiency claim (see Section 3.5.3).

3.9 Summary of the Error Estimation Framework

The proposed error estimation framework provides:

- **A conditional a priori bound** (Theorem 1), valid under explicit well-posedness, regularity, and approximation-capacity assumptions (A1)–(A4).
- **A computable a posteriori indicator** (49), built from pointwise residual evaluation on a verification set, with no artificial mesh-jump terms.
- **A reliability theorem** (upper bound), together with an honest statement that a matching efficiency (lower bound) result is not established here for the meshfree setting.
- **Practical guidelines** for computing and validating the indicator empirically.

Remark 3. *The Dirichlet boundary control setting analyzed here (Assumption (A2)) is a known source of low-regularity difficulties for the classical PDE theory; results here should not be read as removing that difficulty, only as working under regularity hypotheses strong enough to bypass it. Extending the framework to genuinely low-regularity Dirichlet data, and to a provably efficient meshfree a posteriori indicator, are left for future work.*

4 Numerical Experiments

4.1 Common computational settings

Both tests use $\Omega = (0, 1)$, $T = 1$, $\nu = 0.1$, and $\alpha = 10^{-2}$. All networks are fully connected, use tanh activations and Xavier initialization, and are trained in double precision. A single control network $U_{\psi}(x, t)$ is evaluated at $x = 0$ and $x = 1$ to obtain the two boundary traces.

Bảng 1: Neural-network architectures.

Network	Input	Output	Hidden layers	Neurons/layer
Y_{θ} or N_{θ} (state)	(x, t)	scalar	4	50
U_{ψ} (control)	(x, t)	scalar	3	30
M_{ϕ} (adjoint)	(x, t)	scalar	4	50

Training uses independent uniform Monte-Carlo samples: 10 000 interior points, 2 000 points on each boundary component, and 500 initial points. Points are resampled every 500 Adam epochs. Adam is run for 20 000 epochs from learning rate 10^{-3} , with decay factor 0.9 every 1 000 epochs, followed by at most 5 000 L-BFGS iterations. The loss weights are listed in Table 2. The reported values correspond to one run with seed zero.

Bảng 2: Loss weights used in both experiments.

Term	Direct PINN	Indirect PINN
State residual	1	1
State boundary condition	10	hard-constrained
State initial condition	10	10
Adjoint residual	—	1
Stationarity residual	—	1
Adjoint boundary and terminal conditions	—	hard-constrained

Accuracy is reported using the root-mean-square error

$$\text{RMSE}(v) := \left(\frac{1}{N} \sum_{j=1}^N |v_j - v_j^*|^2 \right)^{1/2}. \quad (52)$$

Residual maps are evaluated on a verification grid independent of the training samples.

4.2 Example 1: Manufactured solution from the original formulation

4.2.1 Problem definition

The first test is a cubic reaction–diffusion equation with $f(y) = y^3$. We prescribe

$$y^*(x, t) = \sin(\pi x)e^{-t}, \quad u^*(0, t) = u^*(1, t) = 0, \quad (53)$$

and $y_0(x) = \sin(\pi x)$. The manufactured source is

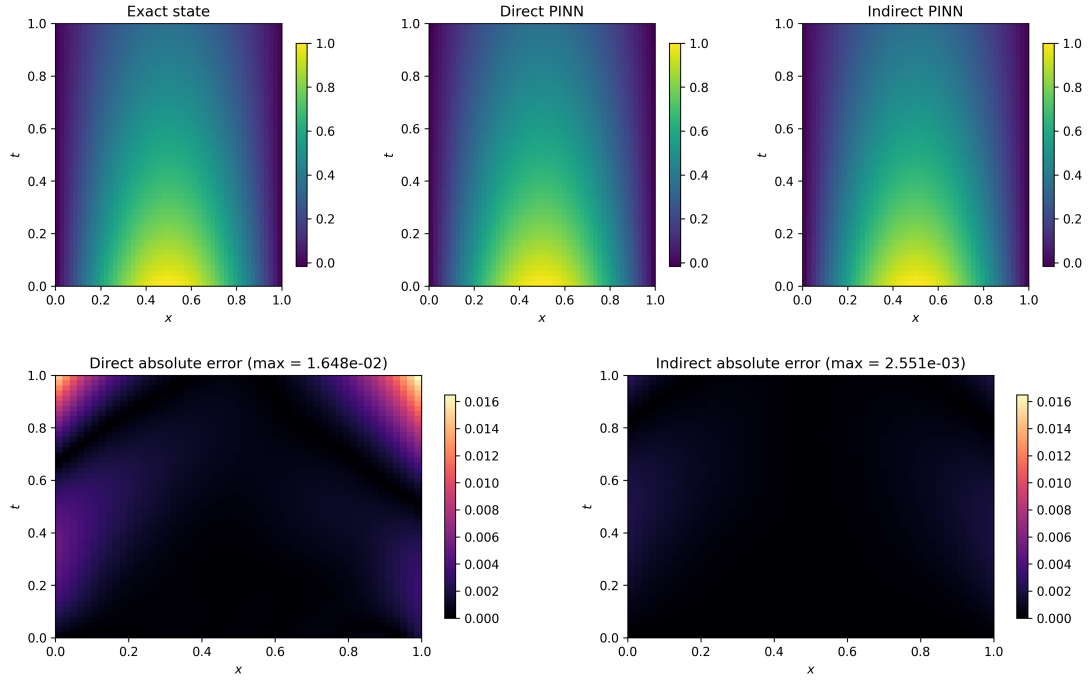
$$g(x, t) = (-1 + \nu\pi^2) \sin(\pi x)e^{-t} + \sin^3(\pi x)e^{-3t}. \quad (54)$$

Choosing $y_d = y^*$ makes (y^*, u^*) feasible with $J(y^*, u^*) = 0$; hence it is a global optimum. The exact adjoint is $\lambda^* = 0$.

4.2.2 Direct and Indirect results

Figure 1 compares the exact state and the two PINN approximations. The direct and indirect state RMSEs are 2.34×10^{-3} and 6.80×10^{-4} , respectively. Their maximum pointwise state errors are 1.648×10^{-2} and 2.551×10^{-3} .

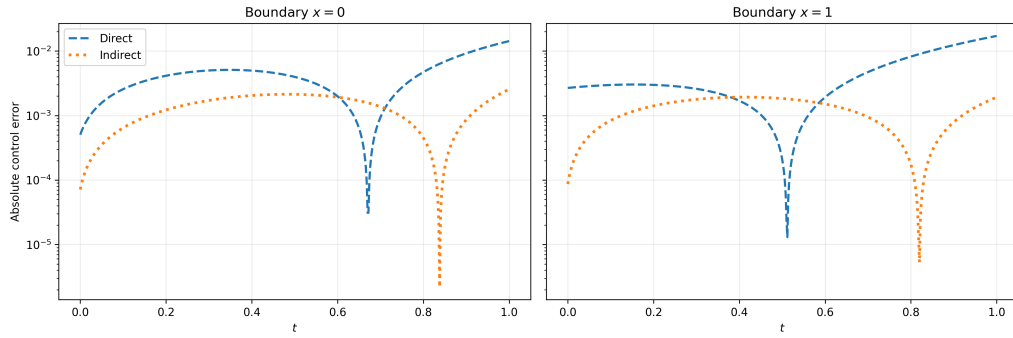
State comparison — Cubic reaction-diffusion test



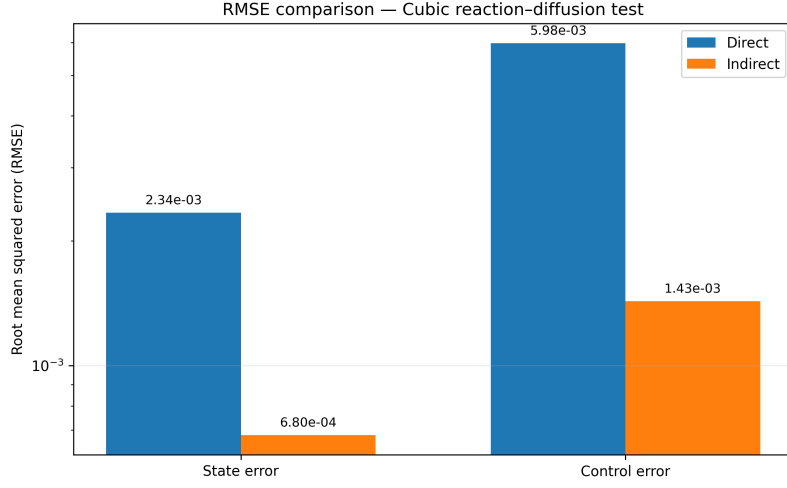
Hình 1: State approximation and absolute error for the cubic reaction-diffusion test. Both error panels use the same color scale.

The boundary-control RMSE decreases from 5.98×10^{-3} for the direct method to 1.43×10^{-3} for the indirect method; see Figures 2–3.

Boundary-control errors — Cubic reaction-diffusion test



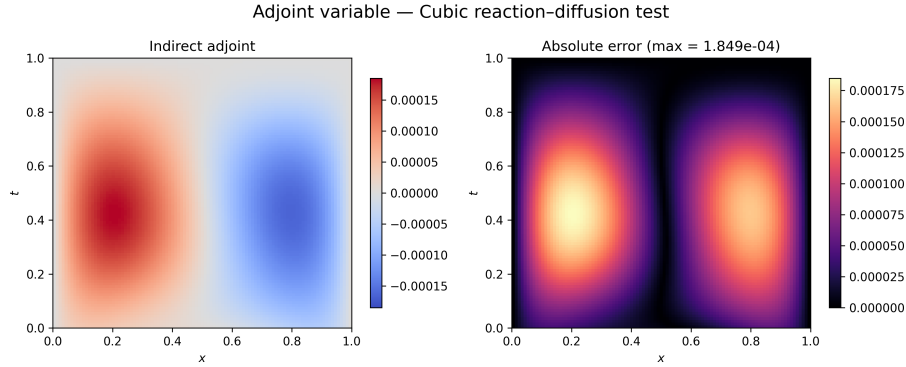
Hình 2: Absolute boundary-control errors on the left and right endpoints for Example 1.



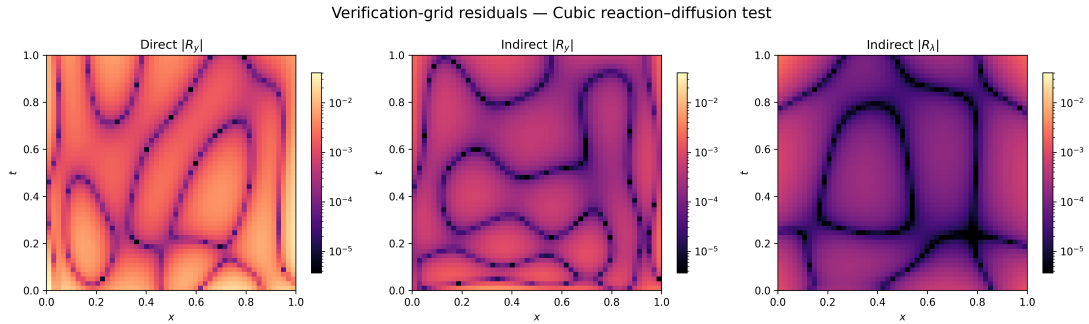
Hình 3: RMSE comparison for Example 1.

4.2.3 Discussion

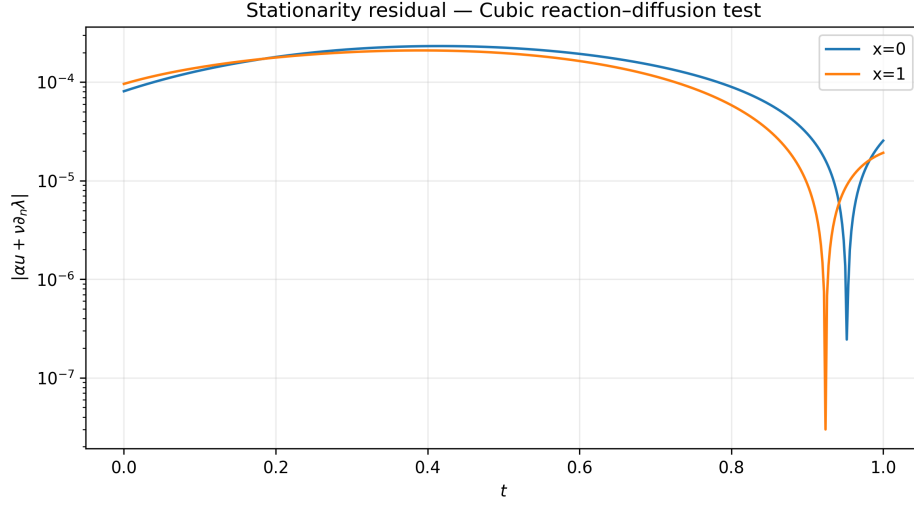
The indirect formulation reduces the state RMSE by approximately 70.9% and the control RMSE by approximately 76.1%. Its largest adjoint error is 1.849×10^{-4} , consistent with the exact value $\lambda^* = 0$ (Figure 4). The verification maps in Figure 5 show generally smaller state residuals for the indirect method. The stationarity residual remains mainly of order 10^{-4} and decreases near the terminal time (Figure 6).



Hình 4: Indirect adjoint approximation and absolute error for Example 1.



Hình 5: State and adjoint residuals evaluated on an independent verification grid for Example 1. A common logarithmic color scale is used.



Hình 6: Stationarity residual $|\alpha u + \nu \partial_n \lambda|$ at both endpoints for Example 1.

4.3 Example 2: Nontrivial manufactured KKT solution

4.3.1 Construction of the exact KKT solution

The second test uses the linear heat equation, $f(y) = 0$, and is constructed so that both the control and adjoint are nonzero. Let $A = 0.05$ and prescribe

$$\lambda^*(x, t) = A x(1 - x) \sin(\pi t), \quad (55)$$

$$u^*(0, t) = u^*(1, t) = \frac{\nu A}{\alpha} \sin(\pi t) = 0.5 \sin(\pi t), \quad (56)$$

$$y^*(x, t) = 0.5 \sin(\pi t) + \sin(\pi x) e^{-t}. \quad (57)$$

The source and desired trajectory are then defined from the state and adjoint equations:

$$g(x, t) = 0.5\pi \cos(\pi t) + (-1 + \nu\pi^2) \sin(\pi x) e^{-t}, \quad (58)$$

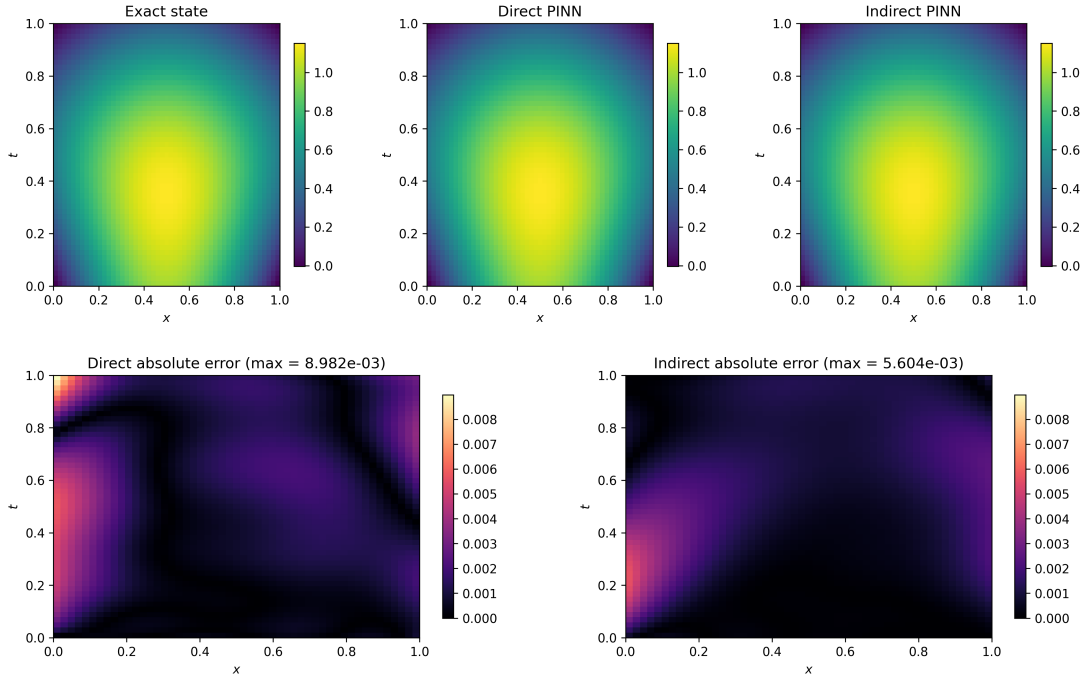
$$y_d(x, t) = y^*(x, t) - A\pi x(1 - x) \cos(\pi t) + 2\nu A \sin(\pi t). \quad (59)$$

The triplet (y^*, u^*, λ^*) satisfies the complete optimality system. Since the dynamics are linear and the objective is strictly convex in the control, it is the unique optimum.

4.3.2 Direct and Indirect results

Figure 7 shows close agreement with the exact state. The direct and indirect state RMSEs are 1.65×10^{-3} and 1.38×10^{-3} ; the corresponding maximum errors are 8.982×10^{-3} and 5.604×10^{-3} .

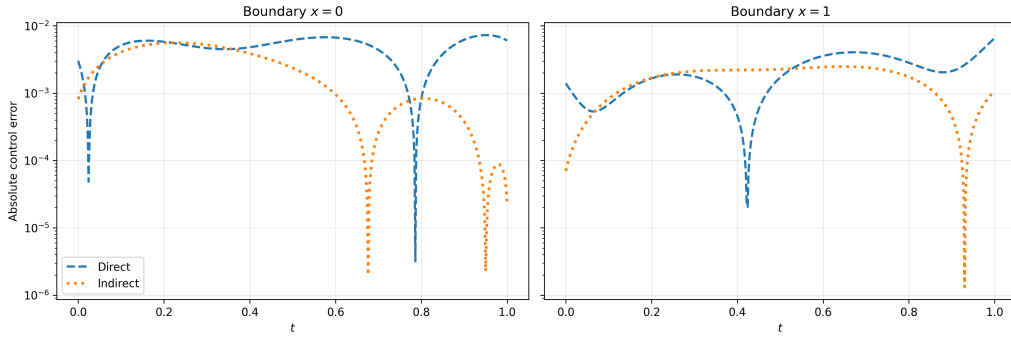
State comparison — Linear boundary-control test



Hình 7: State approximation and absolute error for the nontrivial manufactured KKT test.

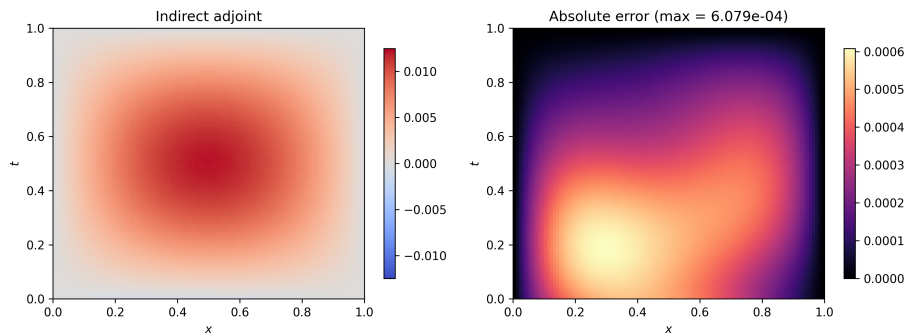
The control RMSE decreases from 4.08×10^{-3} to 2.55×10^{-3} (Figures 8–10). The indirect adjoint has maximum absolute error 6.079×10^{-4} (Figure 9).

Boundary-control errors — Linear boundary-control test

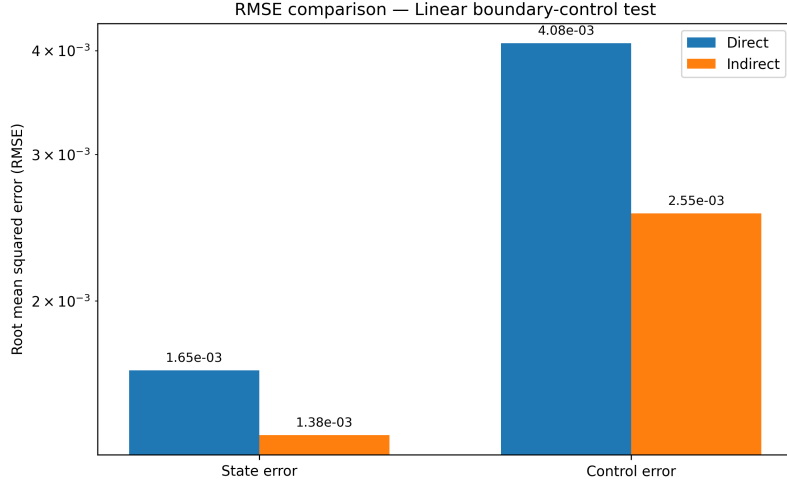


Hình 8: Absolute boundary-control errors for Example 2.

Adjoint variable — Linear boundary-control test



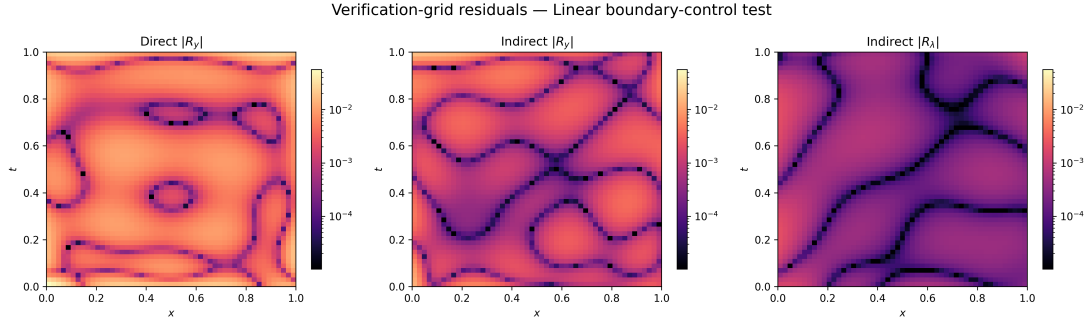
Hình 9: Indirect adjoint approximation and absolute error for Example 2.



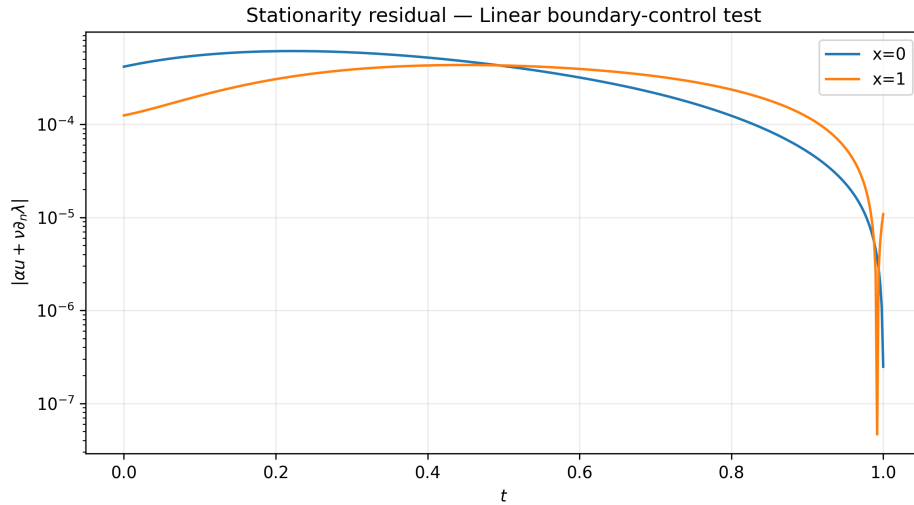
Hình 10: RMSE comparison for Example 2.

4.3.3 Discussion

The indirect method reduces the state RMSE by approximately 16.4% and the control RMSE by approximately 37.5%. Unlike Example 1, this test exercises a nonzero control, a backward adjoint equation, the terminal condition, and the stationarity relation. The control error is not pointwise smaller at every time, but its global RMSE is lower. Figures 11 and 12 show that the learned fields remain consistent with the PDE and first-order optimality conditions on independent verification points.



Hình 11: Verification-grid state and adjoint residuals for Example 2.



Hình 12: Stationarity residual at both endpoints for Example 2.

4.4 Overall comparison

Bảng 3: Summary of the numerical errors.

Example	Method	State RMSE	Control RMSE	Maximum state error
1	Direct	2.34×10^{-3}	5.98×10^{-3}	1.648×10^{-2}
1	Indirect	6.80×10^{-4}	1.43×10^{-3}	2.551×10^{-3}
2	Direct	1.65×10^{-3}	4.08×10^{-3}	8.982×10^{-3}
2	Indirect	1.38×10^{-3}	2.55×10^{-3}	5.604×10^{-3}

For the present runs, the indirect formulation gives lower state and control RMSE in both examples. The improvement is strongest in Example 1, while Example 2 confirms that the method also recovers a nontrivial boundary control and adjoint. These single-seed results indicate an advantage of the hard-constrained optimality-system formulation, but repeated runs are required for a statistical comparison.

5 Conclusions

We developed direct and indirect PINN formulations for a semilinear parabolic optimal-control problem with Dirichlet boundary control. A single network $U_\psi(x, t)$ represents the control on both boundary components. In the indirect method, a boundary lifting enforces the state-control coupling exactly, while the adjoint architecture enforces both its homogeneous boundary condition and terminal condition.

Two manufactured tests were considered. The cubic reaction–diffusion test reproduces the original zero-control setting, whereas the linear KKT test contains a nonzero control and adjoint. In the reported runs, the indirect method produced lower state and control RMSE in both examples. Independent verification-grid residuals also showed consistency with the state, adjoint, and stationarity equations.

The error estimates remain conditional on well-posedness, regularity, stability, and network approximation assumptions. In particular, no efficiency result is claimed for the meshfree residual indicator. Future work should include repeated random seeds, convergence studies with increasing collocation sets, genuinely low-regularity Dirichlet data, and nonlinear tests with nontrivial controls.

Tài liệu

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